

Green Energy and Technology

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Magnetocaloric Energy Conversion

From Theory to Applications

 Springer

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“I believe we are witnessing history in the making.”

Karl A. Gschneidner Jr., At the unveiling of the prototype, proof-of-principle, near room temperature magnetic refrigerator at the Astronautics Technology Center in Madison, Wisconsin, on February 20, 1997.

To those that have been and are working in the field: “The limitations of magnetic refrigeration are only in the minds of the individual engineers and scientists.”

Karl A. Gschneidner Jr.

“Practical magnetocaloric energy conversion on a large scale has been a grand challenge to the ingenuity of engineers and scientists off and on since the time of Edison and Tesla. Recent years has seen the concept brought ever closer to practicality. Until now there has been no comprehensive book on the subject.”

R.E. Rosensweig

Preface

This book provides the basis for engineering a new, alternative form of technology, i.e., magnetocaloric energy conversion. It has been written for postgraduate students, engineers and researchers, especially those working in the areas of refrigeration and heat pumping, power generation and heat transfer. The aim of the book is to provide the reader with a theoretical basis supported by practical examples, facilitate the understanding, design and construction, and to introduce new solutions to engineering problems in future magnetocaloric energy-conversion devices.

The global energy demand for refrigeration and air conditioning is rapidly increasing. Nearly 20 % of all energy is wasted in existing cooling or air-conditioning devices. Here, vapour-compression air-conditioning and refrigeration technologies have dominated the market for more than 100 years.

As the capacity per unit mass of vapour-compression technology has improved over time their dominant position has become even stronger by lowering the manufacturing costs and improving efficiency. However, the trends of the converging S-curve of development show that vapour compression has become a mature technology, having only a small potential for some significant improvements in efficiency.

There is a continuous quest to find new, environmentally friendly refrigerants for vapour compression. Many of the alternative substances are related to a lower energy efficiency, high pressures, flammability or even explosive hazards.

Based on the adopted Kyoto Protocol, the chlorofluorocarbons CFCs, which had an important impact on the ozone layer (with high ozone depletion potential (ODP)), were phased out in the 1990s. Another group of refrigerants, the hydrochlorofluorocarbons (HCFs) will be phased out by 2020. Despite this, the majority of remaining refrigerants possess a global warming potential (GWP). Moreover, most of the electricity production is based on fossil fuels. In order to reduce the amount of their use, one way is certainly to increase the use of renewable energy sources. On the other hand, one should take care of reducing energy consumption throughout the whole energy chain—from production to consumption. The last is also associated with the efficiency of energy conversion.

Therefore, any improvement in the efficiency of energy-conversion technologies will drastically influence the global energy demand as well as reduce the harmful environmental impacts. Vapour compression represents an important part of the world's energy consumption. Since it represents a mature technology, it is therefore obvious to think about alternative technologies. One such example is magnetocaloric energy conversion, i.e., magnetic refrigeration, magnetic heat pumping, and magnetic power generation, respectively.

Magnetocaloric energy conversion is a technology that is based on the exploitation of the magnetocaloric effect (MCE). The MCE is a physical phenomenon that occurs in magnetic materials under the influence of a varying magnetic field, i.e., magnetization and demagnetisation. Instead of a gas-refrigerant as the working substance, magnetocaloric devices employ a solid magnetic material.

Magnetocaloric materials have a GWP and an ODP equal to zero. In analogy with the polytropic compression and isenthalpic expansion of the gas-refrigerant, the processes of magnetization and demagnetization of some magnetocaloric materials are nearly isentropic. This is why the magnetocaloric energy conversion, as an alternative, offers improvements in efficiency. Note that the earliest prototypes have already demonstrated an exergy efficiency higher than that of existing vapour-compression technologies. Moreover, silent operation without vibration makes this technology attractive for a large number of applications. It is therefore not a coincidence that different technology foresights have characterized magnetocaloric energy conversion as one of the most promising alternatives for future air conditioning and refrigeration.

In 1843, James Prescott Joule observed that heat is evolved in iron samples under an applied magnetic field. In 1860, William Thomson (Lord Kelvin) knew that ferromagnetic materials lose their magnetic properties when heated above a certain temperature (known today as the Curie temperature). Twenty years later in 1881 Emil Warburg published his work explaining magnetic hysteresis. A year later, in 1882, James Alfred Ewing discovered the same phenomenon and was the first to name it "hysteresis". Later in 1917–1918, the physicists Pierre Weiss and Auguste Piccard announced the discovery of the "novel magnetocaloric phenomenon".

In parallel with the discovery of the MCE, scientists were creating ideas to apply the MCE in energy-conversion devices. Towards the end of the 19th century Jožef Stefan had the first ideas that ferromagnetic materials could be applied in power generation. Some years later, but still in the 19th century, Nikola Tesla and Thomas Alva Edison patented ideas about thermomagnetic motors. In 1926 Peter Joseph William Debye and in 1927 William Francis Giauque independently discussed the application of the MCE for cryogenic temperatures under 1 K. The experimental proof came in 1933 from W.F. Giauque and D.P. MacDougall. The first magnetic refrigerator ever built for room-temperature applications was constructed and tested by G.V. Brown in 1976. In 1982, John A. Barclay and William A. Steyert introduced the idea of an active magnetic regenerator (AMR), which is today widely applied.

An important milestone in the development of magnetocaloric energy conversion is the discovery of the so-called giant MCE, which was announced in 1997 by Karl A. Gschneidner, Jr. and Vitalij K. Pecharsky. Since then, there has been an exponential increase in patents, articles, conference contributions as well as individual chapters in a number of different books.

Among the books that concern *Magnetocaloric Energy Conversion*, it is important to mention that on *Ferrohydrodynamics* by Ronald E. Rosensweig, which was published in 1985. Although this book could be taken as the bible for ferrohydrodynamics, a large part of it was dedicated to magnetocaloric energy conversion, not only by providing the basic theory, but also by encouraging researchers to enter the engineering of this particular domain.

Another important work was published by Alexander M. Tishin and Yuri I. Spichkin in 2003. The book *The Magnetocaloric Effect and its Applications* was actually the first that was strictly focused on magnetocalorics, by giving a comprehensive description of magnetocaloric materials and their properties, as well as by pointing out different potential applications.

A very large number of different publications on magnetocaloric materials and systems have been provided in the proceedings of the International Conferences on Magnetic Refrigeration at Room Temperature—THERMAG. Since starting in 2005 in Montreux, Switzerland, these conferences have been organized in 2007 in Portorož, Slovenia, in 2009 in Des Moines, Iowa, USA, in 2010 in Baotou, Inner Mongolia, China, in 2012 in Grenoble, France, and in 2014 in Victoria, British Columbia, Canada.

For the continuous organization of the first five THERMAG events under the umbrella of the *IIR—International Institute of Refrigeration*, we have to thank Peter William Egolf, who in 2004 established the *IIR Magnetic Cooling Working Party* and led it until 2012. Without his efforts and also the efforts of all the contributors to such conferences, the magnetocaloric community would be missing an important part, which led to closer collaborations between engineers, physicists and material scientists as well as leading to better transfer of knowledge.

With about 60 prototypes being developed all over the world, and the involvement of different industries, magnetocaloric energy conversion is today on its way towards the first market applications. For these, strong interdisciplinary engineering knowledge is required. The authors of this book have recognised that there exists no such complete work that would provide the required basis for engineering.

The aim of this book is therefore to provide the most important information for students, engineers and researchers to help them understand the physics behind magnetocaloric energy conversion, and to provide sufficient basis, which will serve for the design and construction of future magnetocaloric energy-conversion devices. Moreover, by adding the latest results in research and completing them with new ideas and concepts, this book should be helpful for researchers and their industrial partners in finding and developing new magnetocaloric market applications.

The first seven chapters provide the fundamentals and practical examples on *Magnetocaloric Energy Conversion* and its various sub-domains. Chapters 8 and 9

provide potential solutions to engineering problems supported by new concept proposals, and economic aspects of magnetocaloric energy conversion. The last chapter is dedicated to other, analogous, alternative technologies of solid-state energy conversion, which might also have a potential for future market applications.

In Chap. 1 on the *Thermodynamics of Magnetocaloric Energy Conversion*, the basic magnetocaloric thermodynamic potentials are presented and described. The state of the art gives an overview of the existing theoretical and experimental approaches to magnetocaloric thermodynamic cycles. Design issues related to different thermodynamic refrigeration cycles are shown as well, described through basic thermodynamic equations, as well as definitions of the coefficient of performance (COP), the exergy efficiency and the theoretical maximum cooling capacity.

Chapter 2 on *Magnetocaloric Materials for Freezing, Cooling and Heat-Pump Applications* introduces different magnetocaloric materials in terms of engineering and their applications. Among those, the most promising magnetocaloric materials are highlighted and presented with their basic thermodynamic properties.

Magnetic Field Source represents one of the most important and also most expensive parts of the magnetocaloric device. Therefore, the optimal design of a magnetic field source is crucial for obtaining a cost-effective and energy-efficient device. In this chapter we briefly describe some of the most important issues that relate to magnetic field sources. Despite the fact that different magnetic field sources with respect to their application in magnetocaloric energy conversion can be applied, the main emphasis of Chap. 3 is on permanent magnets and their assemblies. However, brief information is also provided about electric resistive magnets and superconducting magnets, including their cryogenic parts. A review supported by drawings of different permanent-magnet assemblies is given in the chapter, which includes basic information about their characteristics.

Chapter 4 deals with *Active Magnetic Regeneration*. Note, most of the existing prototype devices and studies are based on applying magnetocaloric materials with the principle of active magnetic regeneration (AMR). In the first part of the chapter, the operation of an AMR within different thermodynamic cycles is explained. Comprehensive information about numerical modelling of an AMR is given, supported by a review of this particular research, and by the necessary mathematical models required for modelling. The impact of the operating conditions (mass-flow rate and operating frequency) and geometrical characteristics on AMR are presented. In addition, results on numerical and experimental investigations of the different AMR thermodynamic cycles are shown, and guidelines for future research on AMR thermodynamic cycles are given. The impact of the heat-transfer fluid on the characteristics of the AMR is discussed. Brief information about some processing and manufacturing techniques is provided. The end of the chapter provides brief information on the practical limitations for the application of AMR cycles.

A special case of magnetocaloric energy conversion relates to *Magnetocaloric Fluids*. These can be further divided into ferrofluids (nanofluids) and magnetorheological particle suspensions. In Chap. 5, this book provides important

information on rheologic constitutive models of magnetic fluids, and introduces the basic governing equations, which relate to the thermo-magneto-fluid dynamics of ferrofluids. The review on existing studies and applications of magnetocaloric fluids is given. The chapter provides a discussion and guidelines for the potential design of devices that concern magnetocaloric energy conversion with magnetocaloric fluids. Since other magnetic fluids can also be successfully applied in magnetocaloric energy conversion, their various potential uses are outlined in the chapter.

The existing practical limitations of AMR principles relate to the power density of magnetocaloric devices. This is associated with the number of thermodynamic cycles performed in a unit of time. Therefore, in the past few years, some research groups have started a serious investigation of the potential in other mechanisms, which led them to research on *Special Heat Transfer Mechanisms—Active and Passive Thermal Diodes*. The thermal diode (heat “semiconductor”, thermal switch, heat valve, thermal rectifier) is a physical phenomenon, mechanism or device in which it is possible to manipulate and control the direction of the heat flux and sometimes also the intensity of the heat flux. Chapter 6 presents different kinds of mechanisms and devices that can be applied as thermal diodes. These can be further divided into two main areas: solid-state thermal diodes and microfluidic thermal diodes, respectively. The characteristics of such thermal diodes are compared and presented with respect to different potential applications. A review of existing research with regard to magnetic refrigeration is given and the potential configurations of thermal diodes in magnetic refrigeration as guidelines for future applications are shown.

Chapter 7 provides an *Overview of Existing Magnetocaloric Prototype Devices*. Generally, two different types of such devices have been exploited—linear and rotary devices. Based on information from the literature, as well as due to the help of the magnetocaloric community, the majority of these prototypes are presented in this chapter. The information is supported by drawings, photographs as well as the main operating characteristics. To this we also add contact details of the institutes and people, which may be used for collaboration, potential investments in the technology or industrial development. At the end of the chapter, the summary of all prototypes is given in tables.

As a continuation of Chap. 7, the Chap. 8 on *Design Issues and Future Perspectives for Magnetocaloric Energy Conversion* gives comprehensive information on the different possible designs for particular types of magnetic refrigerators and heat pumps. Whereas some of the configurations were already applied in research, Chap. 8 also provides information about new solutions, which are based on the author’s research experiences and which might be applied in future studies and related devices. In addition, a note on power generation is added, with a brief review of the existing work in this particular domain. By pointing out the most successful design approaches, this chapter also serves as a future guideline on magnetic refrigeration and heat pumping.

The commercialization of the presented technology requires analyses of the *Economic Aspects of the Magnetocaloric Energy Conversion*. In Chap. 9, we address the cost issues that relate to magnetic field sources and magnetocaloric

materials. With supporting information on global market prices, we have also added a simple calculation of the costs for the magnetocaloric refrigerator. This is followed by a review of the different economic studies covering magnetic refrigeration, including some of the ecological aspects that are related to the carbon footprint and lifecycle analyses (LCAs).

The last chapter in this book is dedicated to other *Alternative Solid-State Energy Conversion* technologies. Because of the analogy with magnetocaloric energy conversion, these “ferroic”, solid-state technologies can be applied by engineers having a knowledge of magnetocalorics. The chapter is divided into three sections, which regard the electrocaloric (pyroelectric), barocaloric and elastocaloric energy conversion technologies. In each section, the physical principle behind the technology is presented with an overview of the existing materials and their physical properties. Furthermore, different possibilities for how to design an energy-conversion device using these materials are reviewed (especially for electrocalorics). Since the technology based on these three effects is in an early stage of development, only a few prototypes of energy-conversion devices, as the state of the art, have been presented.

With the presented knowledge and engineering solutions we hope that this book will well serve the reader in exploiting alternative possibilities of energy conversion by learning and designing future magnetocaloric devices for refrigeration, air-conditioning, heating and power generation.

Ljubljana, September 2014

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Finally, we would like to thank Ronald E. Rosensweig, the father of ferrohydrodynamics, for his helpful remarks and encouraging support.

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Chapter 1

The Thermodynamics of Magnetocaloric Energy Conversion

Magnetocaloric energy conversion is a technology based on the exploitation of the magnetocaloric effect (MCE). The MCE is a physical phenomenon that occurs in magnetic materials under the influence of a varying magnetic field. It is usually expressed as the adiabatic temperature change or isothermal total entropy change of a material. In a ferromagnetic material the entropy can be, for instance, related to the magnetic part and the part related to the temperature of the system (e.g. the lattice entropy). In the absence of a magnetic field, the magnetic moments in the material are disordered. If a magnetic field is applied to the material, the magnetic moments will be forced to align in a higher order. As a consequence, the magnetic entropy will decrease. In isentropic (adiabatic) conditions, the total entropy will remain constant. Therefore, the decreased magnetic entropy will manifest itself in an increased lattice entropy. The atoms in the material will start to vibrate more intensively, and as the consequence, the temperature of the magnetic material will increase. The opposite occurs when the magnetic field is removed: the magnetic entropy is increased and the temperature decreases. On this basis, it is possible to create energy conversion cycles by applying different thermodynamic processes.

In this chapter, the basic magnetocaloric thermodynamic potentials are presented and described. The state of the art gives an overview of the existing theoretical and experimental approaches to magnetocaloric thermodynamic cycles. Different magnetic thermodynamic cycles are described. Besides thermodynamic cycles with conventional simple cycles, an important emphasis is placed on thermodynamic cycles that apply active magnetic regeneration (AMR). Since most of the existing devices apply the AMR principle, a whole chapter (Chap. 4) is dedicated to this topic.

1.1 Introduction

The earliest thermodynamic studies of the magnetocaloric effect near or above room temperature began in the 1950s and 1960s. In addition to a number of cryogenic applications, this work was initially focused on the development of heat engines for the generation of useful power. Researchers were investigating different magnetic

power generation thermodynamic cycles and their specific processes. Their work was based on that of Tesla [1] and Edison [2], who had patented ideas on “pyromagnetic generators”. At this time, electric coils were used as the sources of the magnetic field. However, there is no evidence that such devices were ever built.

At the end of 1950s, one of the first thermodynamic analyses of magnetocaloric power generation was presented by Brillouin and Iskenderian [3]. This was soon followed by other reports [4–7]. Whereas most of the early investigations considered magnetocaloric materials in their solid form, in the 1960s there was a lot of interest in the idea of producing magnetic power generators by using magnetocaloric suspensions as the working fluids. Most of this pioneering work was performed by Resler and Rosensweig [8, 9]. Subsequent work in the 1980s considered magnetocaloric power generators based on solid working materials [10–12]. There is no evidence that any real prototype device for power generation has been developed.

With the discovery of the giant magnetocaloric effect in 1997 [13], which was followed by a number of prototypes for magnetic refrigerators, magnetocaloric power generation has again become an interesting topic. Furthermore, a knowledge about many new materials and the possibility of layering these materials (in order to have a potentially larger temperature span) also led to new activities in magnetic power generation [14–22]. Despite this, most of the published work in the field of thermodynamics in recent years has been dedicated to magnetic refrigeration. This is also the reason why the thermodynamic cycles presented in this chapter relate to refrigeration. A discussion of aspects relating to power generation is therefore only given in the Chap. 8.

The earlier investigations that considered particular thermodynamic cycles were performed by Resler and Rosensweig in the 1960s [8, 9], who mostly did work on magnetocaloric fluids. Brown in 1976 [23], then Steyert in 1978 [24], analysed the thermodynamics of solid magnetocaloric refrigerants, with the latter focused on a magnetic Stirling cycle. Kirol and Mills in 1984 and 1985 [25] performed analyses of the thermodynamics of magnetocaloric power generators with solid magnetocaloric materials. Rosensweig in 1985 [26] published a book on ferrohydrodynamics, with a comprehensive description of the thermodynamics and the fluid dynamics of magnetocaloric fluids. Barclay [27] in 1990 investigated magnetocaloric heat pumps.

No systematic approach has been employed by the research community to evaluate the various magnetocaloric thermodynamic cycles. The first record of this can be found in Chen et al. 1992 [28]. They performed a theoretical evaluation based on four different magnetocaloric thermodynamic cycles: Carnot, Brayton, Stirling, and the so-called “ideal regenerative” cycle. In the last of these, the authors proposed a thermodynamic cycle that is similar to the Stirling cycle. However, there is no evidence of any experimental results relating to such a cycle.

Since 1990 there have been a large number of publications related to the basics of magnetocaloric thermodynamics [29–38].

1.2 Heat, Work and the Basic Thermodynamic Relations

The thermodynamics in this chapter relates to the magnetocaloric material as the observed system. Because of this, we deal with the thermodynamics described with the internal magnetic field in the magnetocaloric material. This should not be misunderstood as the external magnetic field, which is related to the magnetic field source (e.g. the magnetic field in the “empty” air gap of a permanent magnet).

One way to understand magnetic energy is to consider it as a form of potential energy. Imagine a rock on a mountain, having a potential energy (a magnetocaloric material in a magnetic field, produced by a permanent magnet or induced by an electrical coil). In order to put the rock on the mountain, work has to be performed on it (work is performed on the magnetocaloric material when it is magnetized; therefore, the magnetocaloric material receives magnetic work from the permanent magnet or electric field source). When the rock is rolled downhill, its potential energy decreases and the rock does work, e.g. through kinetic energy (the magnetocaloric material does work during the demagnetization process). Actually, during the demagnetization the magnetocaloric material will have to be pulled out of the magnetic field. Looking at the magnetocaloric material, which in our case is the observed system, the magnetocaloric material performs the work.

In the following text, we will assume conditions of constant pressure p and volume V for a solid magnetocaloric material. For a simpler presentation we have written all the equations in their one-dimensional form and apply notation for the exact differential for specific work (dw instead of δw) and specific heat (dq instead of δq).

Figure 1.1 shows an example of a thermodynamically closed system, which is analogous to a piston compressing a gas in a cylinder. In such a system, there is no transfer of mass over the system boundaries (the magnetocaloric material represents the system boundaries). In a thermodynamically opened system, however, there is a mass flow of the magnetocaloric material in and out of the system boundaries. For instance, as shown in Fig. 1.2, a fixed system boundary is shown around the magnetocaloric material. According to Fig. 1.2, the magnetocaloric material rotates (flows) through such a boundary. The first law of thermodynamics for a closed thermodynamic system states that:

$$du = \delta q - \delta w \quad (1.1)$$

The internal energy of the magnetocaloric material will increase if heat is added to a system or if work is performed on the magnetocaloric material. According to Fig. 1.1, the work is performed on the magnetocaloric material by moving the material into a magnetic field, produced by a permanent magnet or by the induction of a magnetic field using an electrical coil. Because of this, the magnetocaloric material is magnetized (Fig. 1.1b), and its internal magnetic field increases.

The specific work required to magnetize the magnetocaloric material in a thermodynamically closed system is equal to (see also [26, 37, 39–46]):

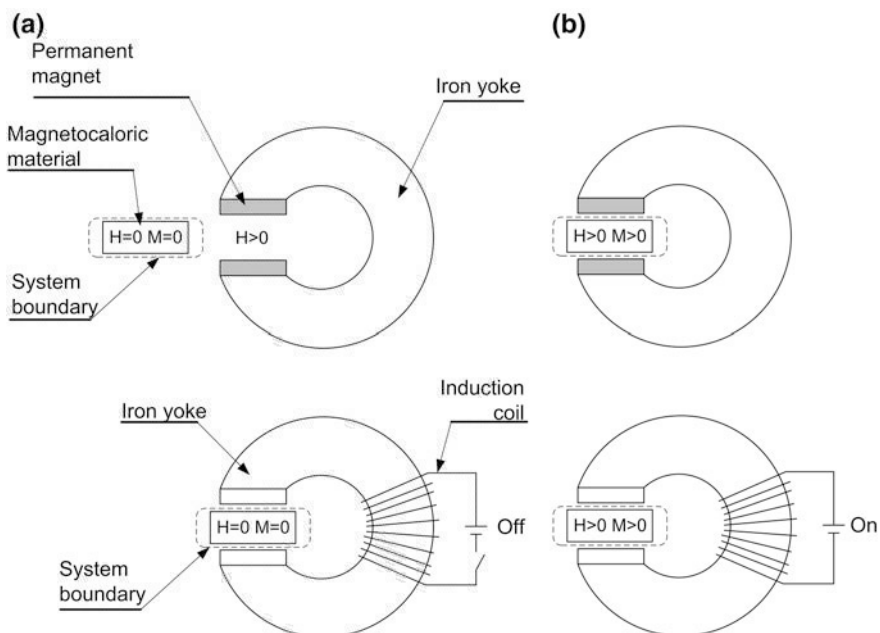


Fig. 1.1 Thermodynamically closed system, **a** Example of a magnetocaloric material in the absence of an external magnetic field (demagnetized state), **b** Example of a magnetocaloric material being magnetized

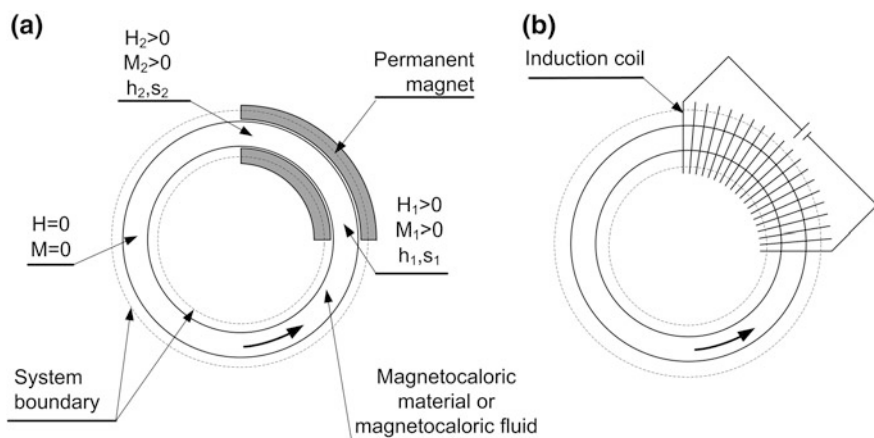


Fig. 1.2 A thermodynamically open system for “flow” of the magnetocaloric material, **a** Example of a magnetocaloric material in the absence of an external magnetic field (demagnetized state) for permanent magnet assembly or electric coil, **b** Example of a magnetocaloric material being magnetized by permanent magnet assembly or electric coil

$$dw = -\mu_0 H dM \quad (1.2)$$

Now, the first law of thermodynamics can be rewritten as:

$$du = dq + \mu_0 H dM \quad (1.3)$$

And for a reversible process, where

$$dq = T ds \quad (1.4)$$

Equation (1.3) takes the following form:

$$du = T ds + \mu_0 H dM \quad (1.5)$$

The derivative of the specific total entropy (for isobaric and isochoric conditions) can be defined in our particular case as:

$$ds(T, H) = \left(\frac{\partial s}{\partial T} \right)_H dT + \left(\frac{\partial s}{\partial H} \right)_T dH \quad (1.6)$$

Applying Eqs. (1.4) and (1.6), we can define the derivative of the specific heat and the specific heat capacities as:

$$dq = c_H(T, H) dT + c_T(T, H) dH \quad (1.7)$$

with:

$$c_H = \left(\frac{\partial q}{\partial T} \right)_H = T \left(\frac{\partial s}{\partial T} \right)_H \quad (1.8)$$

$$c_T = \left(\frac{\partial q}{\partial H} \right)_T = T \left(\frac{\partial s}{\partial H} \right)_T \quad (1.9)$$

The following Maxwell relation can be applied as well:

$$\left(\frac{\partial s}{\partial H} \right)_T = \mu_0 \left(\frac{\partial M}{\partial T} \right)_H \quad (1.10)$$

In an open system (Fig. 1.2) related to the control volume, the magnetocaloric material flows or moves out from the system boundaries. In this particular case, we consider the magnetocaloric fluid or the magnetocaloric solid regenerator to move in such a way (Fig. 1.2).

By applying the Legendre transformation to the first law of thermodynamics (Eq. 1.1), we can derive the definition of enthalpy as:

$$h = u - \mu_0 H M \quad (1.11)$$

Applying Eq. (1.5), the derivative of the specific enthalpy is:

$$dh(s, H) = \left(\frac{\partial h}{\partial s} \right)_H ds + \left(\frac{\partial h}{\partial H} \right)_s dH = T ds - \mu_0 M dH \quad (1.12)$$

In most publications relating to the characterization of magnetocaloric materials, the **isothermal entropy change** is presented as a function of the temperature and the internal magnetic field. It is one of the most widely published properties related to the magnetocaloric effect. The entropy change in an isothermal process can be defined using Eqs. (1.6–1.10) and written as follows:

$$ds(T, H) = \left(\frac{\partial s}{\partial H} \right)_T dH = \mu_0 \left(\frac{\partial M}{\partial T} \right)_H dH \quad (1.13)$$

For a certain increase (or decrease) in the magnetic field between the two states of different magnetic fields under isothermal conditions, the **isothermal entropy change** is defined as follows:

$$\Delta s = s_2 - s_1 = \int_{H_1}^{H_2} \left(\frac{\partial s}{\partial H} \right)_T dH \quad (1.14)$$

$$\Delta s = \int_{H_1}^{H_2} \mu_0 \left(\frac{\partial M}{\partial T} \right)_H dH \quad (1.15)$$

$$\Delta s = \int_{H_1}^{H_2} \frac{c_T}{T} dH \quad (1.16)$$

Another important parameter that is often used for the characterization of magnetocaloric materials is the **adiabatic temperature change**. It denotes the increase or decrease in the temperature due to the increase or decrease of the magnetic field in the absence of a heat flow (adiabatic-isentropic magnetization or demagnetization). In the adiabatic-isentropic process, the total specific entropy does not alter ($ds = 0$). From Eqs. (1.6) and (1.9), it follows that:

$$\left(\frac{\partial s}{\partial T}\right)_H dT = -\left(\frac{\partial s}{\partial H}\right)_T dH = -\mu_0 \left(\frac{\partial M}{\partial T}\right)_H dH \quad (1.17)$$

By using Eq. (1.17) and the definition of the specific heat at a constant magnetic field from Eq. (1.7), it follows:

$$dT = -\frac{\mu_0}{c_H} \cdot T \cdot \left(\frac{\partial M}{\partial T}\right)_H dH \quad (1.18)$$

From Eq. (1.14), we can define the adiabatic temperature change. However, since the measurement of c_H has a much longer characteristic time than the measurement of the magnetization M [46], this kind of calculation is rather problematic. Therefore, the adiabatic temperature change can simply be derived or by using the following equation and knowledge on two temperatures at constant total entropy.

$$\Delta T = \int_{T_1}^{T_2} dT = T_2(s, H_2) - T_1(s, H_1) \quad (1.19)$$

The derivative of the enthalpy in the adiabatic–isentropic case equals:

$$dh = -\mu_0 M dH \quad (1.20)$$

We will further denote the process of heat transfer during a constant magnetic field as the **isofield process of cooling or heating**. Such a process is analogous to isobaric heating and cooling and similar to condensation and evaporation, respectively. However, the temperature of the solid magnetocaloric material is not held constant in such a process.

The derivative of the heat and enthalpy in this case equals (Eqs. 1.8 and 1.12):

$$dq = dh = c_H dT = T \left(\frac{\partial s}{\partial T}\right)_H dT \quad (1.21)$$

Figure 1.3 shows the magnetic field–enthalpy–temperature-specific entropy diagram for a reference magnetocaloric material, i.e. gadolinium. The diagram was constructed using the mean-field approximation [47–49]. This led to information about the specific total entropy as a function of the temperature and the internal magnetic field. With the use of Eq. (1.12), the enthalpy values can be defined. It should be pointed out that the specific enthalpy and entropy presented in Fig. 1.3 are given per unit of mass and not per unit of volume. Note, also, that in analogy with conventional vapour compression this diagram is analogous to a p – h (log p – h diagram), which is usually applied for conventional refrigerants.

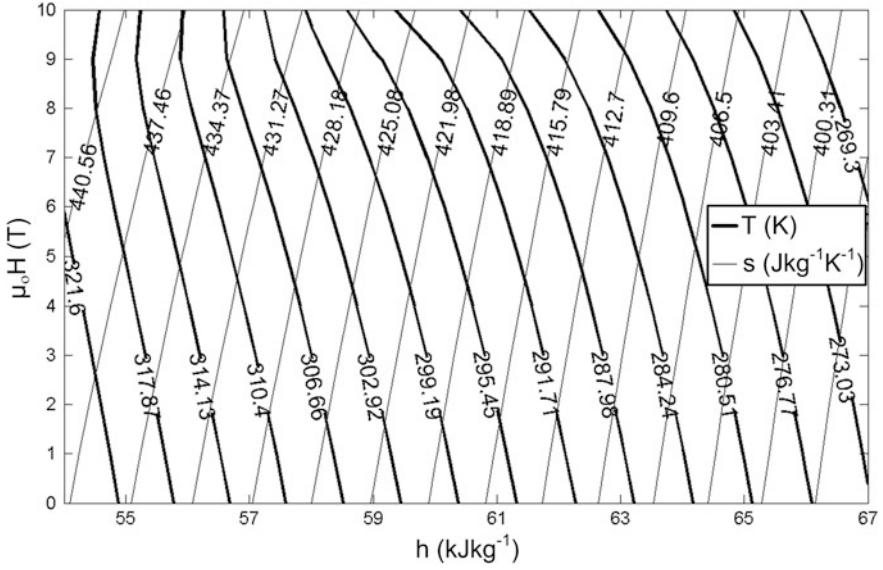


Fig. 1.3 H - h - T - s diagram for the magnetocaloric material gadolinium for magnetic fields between 0 and 10 T

1.3 Magnetocaloric Thermodynamic Cycles

We will start here with some basic equations that relate to cyclic thermodynamics. We will also represent basic magnetic thermodynamic cycles without regeneration.

In any thermodynamic cycle, the cyclic integral of the state function equals zero. This holds true for specific enthalpy, the internal energy, as well as for the magnetic “flow” work denoted by product ($M \cdot H$).

Since the internal energy represents the state function, then Eq. (1.1) for a thermodynamic cycle can be rewritten in the following form:

$$\oint dw = \oint dq = \oint T ds \quad (1.22)$$

Since the cyclic (net) work in the closed system is equal to:

$$\oint dw = -\mu_0 \oint H dM \quad (1.23)$$

And the cyclic integral of the state function, the product of the magnetization and the magnetic field ($M H$), equals zero:

$$\oint d(MH) = \oint M dH + \oint H dM = 0 \quad (1.24)$$

then the work of the closed thermodynamic system equals the work of an open thermodynamic system.

$$\oint dw = -\mu_0 \oint H dM = \mu_0 \oint M dH = \oint dw_t \quad (1.25)$$

By using the character w_t we keep in mind the technical work.

1.3.1 The Coefficient of Performance (COP) and Exergy Efficiency

The ratio between the cooling capacity q_R of the refrigerator and the work w is defined as the coefficient of performance (COP). This can also be expressed in terms of power:

$$\text{COP} = \frac{q_R}{|w|} \quad \text{COP} = \frac{\dot{Q}_R}{P} \quad (1.26)$$

The COP of the ideal Carnot cycle operating at the refrigeration temperature T_R and the heat rejection at ambient temperature can be defined as follows:

$$\text{COP}_c = \frac{T_R}{|T_R - T_{\text{amb}}|} \quad (1.27)$$

Sometimes, the thermodynamic cycles are theoretically “Carnotized” in order to compare such cycles with the COP of the Carnot cycle. Equation (1.27) can be assumed for the “Carnotization” of the Ericsson, Stirling or Rankine cycles. However in other cycles, where the cooling temperature of the refrigerant varies (as is the case in, e.g. the Brayton cycle), the “Carnotization” requires an average value of T_R . In this particular case, we define the average refrigeration temperature as [41]:

$$\overline{T_R} = \frac{q_R}{\Delta s_R} = \frac{q_R}{\int_a^b \frac{dq}{T_R}} \quad (1.28)$$

and the COP of the theoretically Carnotized cycle as:

$$\text{COP}_c = \frac{\overline{T_R}}{|\overline{T_R} - T_{\text{amb}}|} \quad (1.29)$$

The exergy concept is well known and often applied by engineers in different domains. The term *exergy* was suggested by Rant in 1956 [50]. According to Rant [50], energy can be divided into two parts, from which the part that is fully transformable (available) into other kinds of energy is named the “exergy”, and the part that is not transformable (e.g. the internal energy of the ambient) is called the “anergy”. There have been just a few publications on exergy analyses for magnetocaloric energy conversion [21, 51–54].

The exergy efficiency of the magnetic refrigerator represents the ratio between the exergy of the cooling energy and the work (i.e. electric energy, which is pure exergy).

$$\xi = \frac{|E_c|}{W} = \frac{\Delta e_R}{w} \quad (1.30)$$

The specific exergy of the cooling energy can be defined by the following relation (see also Eq. (1.12)):

$$de = (dh - T_{\text{amb}} ds) \quad (1.31)$$

In the case of a constant magnetic field (i.e. the Brayton process, where the temperature T_R is not constant), the enthalpy is equal to the heat (exergy of heat), and therefore:

$$de = (T_R - T_{\text{amb}}) ds = \frac{(T_R - T_{\text{amb}})}{T_R} dq \quad (1.32)$$

and the integration of the exergy between the states a and b (which, e.g. correspond to the cooling process) corresponds to:

$$\Delta e_R = \int_a^b de = \int_a^b \left(1 - \frac{T_{\text{amb}}}{T_R}\right) dq = q_R - T_{\text{amb}} \Delta s_R \quad (1.33)$$

where the second part of Eq. (1.33) represents the anergy. The exergy efficiency can be defined as:

$$\xi = \frac{q_R - T_{\text{amb}} \Delta s_R}{q_H - q_R} \quad (1.34)$$

In the Carnot cycle, the Ericsson cycle or the Stirling cycle, in the case of an **isothermal demagnetization**, the specific exergy of the cooling energy can be defined as: